Model-Guided Deep Hyperspectral Image Super-Resolution

Weisheng Dong, *Member, IEEE*, Chen Zhou, Fangfang Wu, Jinjian Wu, *Member, IEEE*, Guangming Shi, *Fellow, IEEE*, and Xin Li, *Fellow, IEEE*

***Abstract*— The trade-off between spatial and spectral resolution is one of the fundamental issues in hyperspectral images (HSI). Given the challenges of directly acquiring high-resolution hyperspectral images (HR-HSI), a compromised solution is to fuse a pair of images: one has high-resolution (HR) in the spatial domain but low-resolution (LR) in spectral-domain and the other vice versa. Model-based image fusion methods including pan-sharpening aim at reconstructing HR-HSI by solving manually designed objective functions. However, such hand-crafted prior often leads to inevitable performance degradation due to a lack of end-to-end optimization. Although several deep learning-based methods have been proposed for hyperspectral pan-sharpening, HR-HSI related domain knowledge has not been fully exploited, leaving room for further improvement. In this paper, we propose an iterative Hyperspectral Image Super-Resolution (HSISR) algorithm based on a deep HSI denoiser to leverage both domain knowledge likelihood and deep image prior. By taking the observation matrix of HSI into account during the end-to-end optimization, we show how to unfold an iterative HSISR algorithm into a novel model-guided deep convolutional network (MoG-DCN). The representation of the observation matrix by subnetworks also allows the unfolded deep HSISR network to work with different HSI situations, which enhances the flexibility of MoG-DCN. Extensive experimental results are reported to demonstrate that the proposed MoG-DCN outperforms several leading HSISR methods in terms of both implementation cost and visual quality. The code is available at https://see.xidian.edu.cn/faculty/wsdong/Projects/MoG-DCN.htm.**

***Index Terms*— Hyperspectral image super-resolution, model-guided network design, image fusion, deep convolutional network.**

I. Introduction

HYPERSPECTRAL image (HSI) containing a large number of spectral bands has advantages over commonly used multispectral images (MSI) such as RGB data when identifying the physical properties of object materials [1]. Accordingly, HSI has found many successful applications in computer vision from image segmentation [2] and object recognition [3] to image classification [4] and visual tracking [5]–[7]. The trade-off between spatial and spectral resolution has remained one of the great challenges in the practice of HS imaging. Due to various hardware and budget constraints, it is difficult to directly acquire images that have high-resolution (HR) in both spatial and spectral domains. Accordingly, computational approaches to improve the quality of low-resolution (LR) images for HS imaging, such as super-resolution (SR) and pan-sharpening [8], [9] have attracted increasingly more attention. For example, the class of HS Imaging Super-Resolution (HSISR) methods aim at obtaining a HR-HS image by fusing a LR-HSI with a HR-MSI [10].

Existing HSISR methods can be roughly classified into two categories - i.e., *model*-based and *learning*-based. Model-based HSISR methods are mostly developed based upon the linear observation model between the observed images and the original HR-HS images. Despite their theoretical appeal, model-based methods suffer from the hand-crafted prior of HS images, which cannot support end-to-end optimization of HSISR algorithms resulting in limited performance. Accordingly, deep learning -based approaches toward HSISR have attracted increasingly more attention in recent years [10]–[13]. Although these methods often outperform conventional model-based approaches in terms of performance, they lack transparency (i.e., the black-box design) and often suffer from poor generalization property (e.g., from simulation scenarios to real-world applications). How to combine the strengths of model-based and learning-based HSISR has remained an open problem as far as we know.

The motivation behind our approach is largely two-fold. On the one hand, the rich literature in model-based HSISR offers plenty of inspiration for incorporating the domain knowledge (e.g., linear observation model and HS-HR image prior [14]) into network design. Although several model-based methods involve a class of nonconvex regularization, they do not pose additional difficulties to deep learning-based approaches. The training of deep neural networks (DNN) has a natural and intrinsic connection with the optimization of cost functions by numerical method (e.g., gradient descent method [15]). Therefore, it is plausible to translate the existing models developed for HSISR into novel loss functions for deep convolutional network (DCN). On the other hand, there has been a flurry of work on unfolding analytical sparsity

Manuscript received August 12, 2020; revised February 23, 2021; accepted April 21, 2021. Date of publication May 12, 2021; date of current version June 23, 2021. This work was supported in part by the National Key Research and Development Program of China under Grant 2018AAA0101400; and in part by the Natural Science Foundation of China under Grant 61991451, Grant 61632019, Grant 61621005, and Grant 61836008. The work of Xin Li was supported in part by NSF under Grant IIS-1951504 and Grant OAC-1940855, in part by the Department of Justice/National Institute of Justice (DoJ/NIJ) under Grant NIJ 2018-75-CX-0032, and in part by the WV Higher Education Policy Commission under Grant HEPC.dsr.18.5. The associate editor coordinating the review of this manuscript and approving it for publication was Prof. Lisimachos P. Kondi. *(Corresponding author: Fangfang Wu.)*

Weisheng Dong, Chen Zhou, Fangfang Wu, Jinjian Wu, and Guangming Shi are with the School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: wsdong@mail.xidian.edu.cn; 1786715287@qq.com; ffwu_xd@163.com; jinjian.wu@mail.xidian.edu.cn; gmshi@xidian.edu.cn).

Xin Li is with the Lane Department of Computer Science and Electrical Engineering, West Virginia University, Morgantown, WV 26506 USA (e-mail: xin.li@ieee.org).

Digital Object Identifier 10.1109/TIP.2021.3078058

models for network implementation (e.g., [16]–[23]). Unlike conventional wisdom, treating DCN like a black box, such model-guided design leads to transparent network design that is well aligned with the emerging framework of interpretable machine learning [24].

In this paper, we propose to connect these two lines of research by presenting a Model-guided DCN (MoG-DCN) approach to HSISR. To fully exploit the prior knowledge of HSI, we first construct a devoted sub-network for approximating the degradation matrix. Then we leverage recently developed DCN-based image regularization [21] for HSISR and derive an iterative optimization algorithm using the proximal gradient descent algorithm [25]. Based on the approximation of the degradation matrix, a model-guided deep convolutional network (MoG-DCN) is constructed for HSISR by unfolding the proposed iterative optimization algorithm. To further optimize the constructed MoG-DCN in an end-to-end manner, we have paid special attention to the design of reconstruction and denoising modules - i.e., the former acts as the back-projection operator [10] and the latter is exactly the proximal operator [26]. Unlike the single convolutional layer used in [10], our reconstruction module consists of multiple layers enabling superior performance in learning different degradation operators, which further enhances the flexibility of MoG-DCN. Experimental results have shown that the proposed MoG-DCN outperforms many recently proposed methods on both non-blind and blind HSISR.

The rest of the paper is organized below. In Sec. II, we briefly review existing works on model-based and learning-based HSISR. In Sec. III, we proposed an iterative HSISR algorithm based on a DCN regularization. In Sec. IV we present the proposed MoG-DCN by unfolding the model-based iterative reconstruction algorithm. In Sec. V, we report our experimental results and compare them against several other competing approaches, and draw the conclusions in Sec. VI.

II. Related Works

A. Model-Based HSISR

Early works include linear transformation based such as principle component analysis and wavelet transform [27]–[29]. The joint bilateral filtering with the HR-MSI as the guided image has also been proposed [30]. In [31], [32], nonnegative matrix factorization methods were used to reconstruction the HR-HSI from a pair of LR-HSI and HR-MSI. By modeling each HR-HSI patch as a 3D tensor, nonlocal sparse tensor factorization method [33] was proposed for HSISR, where the HSISR problem was formulated as the estimation of the sparse core tensors and dictionaries. As the number of measurements from observation data is much smaller than the unknowns, the recovery of HR-HSI is an ill-posed inverse problem. To improve the accuracy of the estimation, prior knowledge of HS images has been incorporated into the recovery process. In [14], [34]–[36], the pixels of unknown HR-HSI were represented by a linear combination of a few reflectance spectra bases, which were estimated from the observed LR-HSI. To exploit the sparsity prior of HS images, nonnegative sparse representations have been proposed to jointly learn the reflectance spectrum bases and the sparse codes [14], [34]. Considering that the neighboring pixels of HSI have strong correlations, structured sparse coding techniques have also been employed for HSISR [14]. In [37], the self-expressive property of HS image patches within the same cluster was used to regularize the HSISR problem, where the image patches of HR-MS image were used to learn the clusters. To jointly exploit the spatial, spectral, and nonlocal self-similarity correlations, the low tensor-train rank prior of the clustered 3D HSI patches was adopted to regularize the solution of HSISR [38]. By solving the carefully designed objective functions, state-of-the-art HSISR results have been achieved [14], [37], [38].

B. Deep Learning-Based HSISR

Inspired by the great success of deep learning for various computer vision tasks [39]–[42], inverse problems [17], [19], [22], and image restoration applications [21], [43]–[46], deep convolutional networks have also been applied to HSISR [10]–[13]. In [11], 3D-CNN has been proposed for HSISR, where the initially interpolated LR-HS image and the HR-MS image were stacked as the input to the network. Through 3D convolutional layers, the network outputs the HR-HS images. Through end-to-end training, state-of-the-art HSISR performances have been achieved by deep convolutional networks. Despite their excellent performance, these HSISR methods based on deep convolutional network (DCN) have not exploited the domain knowledge encoded into the HSI observation model. To overcome this drawback, a fusion net was constructed in [13] to jointly exploit the observation model and the low-rank constraints. However, in [13] the DCN denoiser was used to regularize the intermediate HR-MS images, leaving room for further improvement. In [10], a DCN was developed based on the back-projection technique for blind HSISR, where the DCN denoiser was used to extract the spatial and spectral details from the residual error to refine the HR-HS image estimates.

III. Model-Based HSI Super-Resolution Algorithm

A. The Objective Function

Let $\boldsymbol{Z} \in \mathbb{R}^{L\times N}$ denote the HR-HSI to be recovered from a pair of LR-HSI $\boldsymbol{X} \in \mathbb{R}^{L\times n}$ and HR-MSI $\boldsymbol{Y} \in \mathbb{R}^{l\times N}$, where L and l denotes the number of spectral bands of the HSI and the MSI, and $N = W{*}H$ and $n = w{*}h$ ($h \ll H$, $w \ll W$) denotes the number of pixels of the HR-MSI and LR-HSI, respectively. The linear relationship between the observed images and the original HR-HSI can be written by [14]

$$\boldsymbol{X}_{L\times n} = \boldsymbol{Z}_{L\times N}\boldsymbol{H}_{N\times n},\ \boldsymbol{Y}_{l\times N} = \boldsymbol{P}_{l\times L}\boldsymbol{Z}_{L\times N}, \qquad (1)$$

where $\boldsymbol{H}$ is the *spatial* downsampling matrix sized by $N \times n$ consisting of a blurring operator followed by subsampling operator and $\boldsymbol{P}$ denotes the *spectral* downsampling matrix sized by $l \times L$.

Since the recovery of the HR-HSI from a pair of HR-MSI and LR-HSI is an ill-posed inverse problem, regularization techniques encoding the prior knowledge of HSI are often required. Generally speaking, the objective function for HSISR can be formulated as

$$\hat{\boldsymbol{Z}} = \underset{\boldsymbol{Z}}{\operatorname{argmin}} \frac{1}{2}||\boldsymbol{X} - \boldsymbol{Z}\boldsymbol{H}||_F^2 + \frac{1}{2}||\boldsymbol{Y} - \boldsymbol{P}\boldsymbol{Z}||_F^2 + J(\boldsymbol{Z}), \quad (2)$$

where $J(\boldsymbol{Z})$ is a regularization term reflecting the prior knowledge of the HSI. Different regularization terms have been proposed for this inverse problem - e.g., the non-negative ℓ_1 sparsity regularization [34], the structured ℓ_1 sparsity regularization [14], and the nonlocal self-expressive prior [37]. Though promising results have been achieved, the optimality of these hand-crafted priors is often questionable. In this work, we propose to regularize this inverse problem with the regularization *implicitly* learned from a set of training HSIs. Inspired by the great success of deep convolutional network (DCN) for various low-level image processing tasks [21], [43]–[46], we propose to use the DCN as a denoising prior for HSISR, which cannot be expressed explicitly.

B. The Proximal Gradient Descent Algorithm

For the objective function of Eq. (2) with the DCN denoising prior, we use the efficient Proximal Gradient Descent (PGD) algorithm [26] to solve Eq. (2), which contains a differentiable part and a non-differentiable part. Let $F(\boldsymbol{Z}) = f(\boldsymbol{Z}) + J(\boldsymbol{Z})$, where $f(\boldsymbol{Z})$ and $J(\boldsymbol{Z})$ denote the differentiable and non-differentiable parts, respectively. As a generalization of the gradient descent, the PGD algorithm minimizes the function $F(\boldsymbol{Z})$ by iteratively computing

$$\boldsymbol{Z}^{(t+1)} = \text{Prox}_{\lambda J}(\boldsymbol{Z}^{(t)} - \lambda \nabla_{\boldsymbol{Z}} f(\boldsymbol{Z}^{(t)})), \quad (3)$$

where $\text{Prox}(\cdot)$ denotes the proximal operator related to the non-smooth term $J(\boldsymbol{Z})$ and λ is the step size. When $f(\boldsymbol{Z})$ is Lipschitz continuous, the step size λ can be set as $\lambda = 1/L$, where L is the Lipschitz constant of $\nabla_{\boldsymbol{Z}} f$. If L is unknown, the step size $\lambda = \lambda^{(t)}$ can also be set adaptively in each iteration by a line search method. In each iteration of Eq. (3), we first minimize the differentiable part $f(\boldsymbol{Z})$ by a gradient descent, followed by the proximal operation. The $\text{Prox}_{\lambda J}(\cdot)$ operator of function $J(\boldsymbol{Z})$ can be defined as

$$\text{Prox}_{\lambda J}(\boldsymbol{V}) = \underset{\boldsymbol{Z}}{\operatorname{argmin}} \frac{1}{2}||\boldsymbol{V} - \boldsymbol{Z}||_F^2 + \lambda J(\boldsymbol{Z}). \quad (4)$$

The $\text{Prox}_{\lambda J}(\cdot)$ operator can be implemented by a Gaussian denoising algorithm applied to the noisy observation $\boldsymbol{V}$. The Gaussian denoiser $g(\cdot)$ associated with the regularizer $J(\boldsymbol{Z})$, can either be the conventional model-based denoiser that can be explicitly expressed in a closed-form, or be more sophisticated learning-based denoiser that was obtained by a DCN training procedure.

Applying the PGD algorithm to the objective function of Eq. (2), $f(\boldsymbol{Z})$ corresponds to the first two fidelity terms, i.e.,

$$f(\boldsymbol{Z}) = \frac{1}{2}||\boldsymbol{X} - \boldsymbol{Z}\boldsymbol{H}||_F^2 + \frac{1}{2}||\boldsymbol{Y} - \boldsymbol{P}\boldsymbol{Z}||_F^2, \quad (5)$$

and $J(\boldsymbol{Z})$ is the regularizer. The gradient of $f(\boldsymbol{Z})$ can be easily computed as

$$\nabla_{\boldsymbol{Z}} f(\boldsymbol{Z}) = (\boldsymbol{Z}^{(t)}\boldsymbol{H} - \boldsymbol{X})\boldsymbol{H}^\top + \boldsymbol{P}^\top(\boldsymbol{P}\boldsymbol{Z}^{(t)} - \boldsymbol{Y}), \quad (6)$$

Algorithm 1 Proposed Iterative HSISR Algorithm

- **Initialization**:
  - (1) Set λ and $t = 0$;
  - (2) Initialize $\boldsymbol{Z}$ as $\boldsymbol{Z}^{(0)}$ with bicubic interpolation;
- **While** not converge **do**
  - (3) Compute $\boldsymbol{V}^{(t)}$ with Eq. (7)
  - (4) Compute $\boldsymbol{Z}^{(t+1)} = g(\boldsymbol{V}^{(t)})$;
  - (5) $t = t + 1$.

End while

where the right-multiplying $\boldsymbol{H}\boldsymbol{H}^\top$ and left-multiplying $\boldsymbol{P}^\top\boldsymbol{P}$ are $N \times N$ and $L \times L$ square matrices respectively. Therefore, in each iteration, we first compute an intermediate estimate $\boldsymbol{V}^{(t)}$ of the original HR HSI as

$$\boldsymbol{V}^{(t)} = \boldsymbol{Z}^{(t)} - \lambda((\boldsymbol{Z}^{(t)}\boldsymbol{H} - \boldsymbol{X})\boldsymbol{H}^\top + \boldsymbol{P}^\top(\boldsymbol{P}\boldsymbol{Z}^{(t)} - \boldsymbol{Y})), \quad (7)$$

followed by the proximal operator $\text{Prox}_{\lambda J}(\cdot)$ applied to $\boldsymbol{V}^{(t)}$- i.e.,

$$\boldsymbol{Z}^{(t+1)} = \text{Prox}_{\lambda J}(\boldsymbol{V}^{(t)}) = \underset{\boldsymbol{Z}}{\operatorname{argmin}} \frac{1}{2}||\boldsymbol{V}^{(t)} - \boldsymbol{Z}||_F^2 + \lambda J(\boldsymbol{Z}). \quad (8)$$

In summary, the iterative algorithm for solving the HSISR problem of Eq. (2) is given in **Algorithm 1**, where we initialize $\boldsymbol{Z}^{(0)}$ with a bicubic interpolated version of $\boldsymbol{X}$. Under the framework of model-based HSISR, the PGD algorithm usually requires dozens of iterations to converge. Note that in **Algorithm 1** any existing HSI denoiser $g(\cdot)$ can be used to compute $\boldsymbol{Z}^{(t+1)}$. Due to the excellent denoising capability of DCN for natural images, we propose to use a DCN denoiser to approximate the proximal operator. Specifically, a U-net like DCN that can exploit the multi-scale dependencies of HSI is employed. The use of DCN-based denoisers also facilitates the end-to-end training of the whole network after model-guided unfolding, as will be elaborated in the next section.

IV. Model-Guided Deep Convolutional Network

In **Algorithm 1**, we have presented a general HSISR algorithm, where the step size λ has to be selected by hand and the denoiser $g(\boldsymbol{Z})$ has to be optimized on a Gaussian denoising task for a specific noise level. Like other model-based image restoration, it is difficult to optimize the regularization parameter λ and functional $g(\boldsymbol{Z})$ jointly, especially due to the nonlinearity of regularization term. An alternative approach is to unfold the iterative optimization algorithm into a series of network implementations as demonstrated by recent works [18], [20], [21], [23]. The total number of unfolding stages naturally corresponds to that of PGD iterations. Through end-to-end training, the network parameters of the denoiser and other parameters of the HSI reconstruction algorithm can be jointly optimized. Note that even the regularization parameter λ was implicitly optimized in the learned denoisers, which differs this work from the conventional wisdom of PGD algorithm [26]. Under the context of HSISR, both the updating step (i.e., the differential updating step consisting of

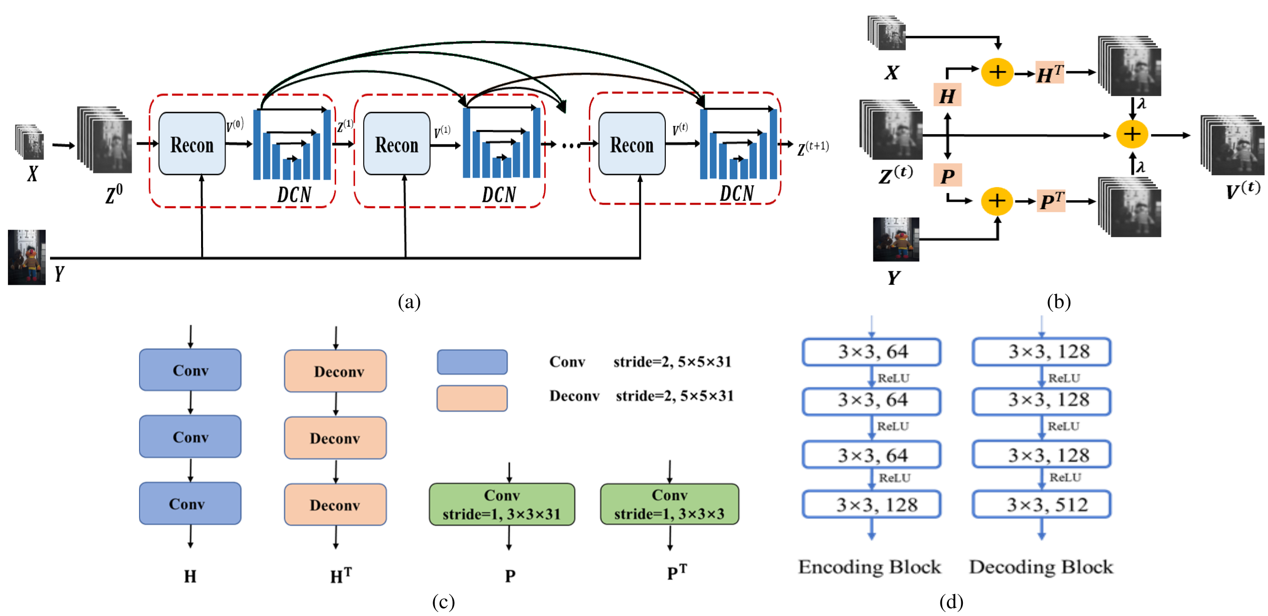


Fig. 1. Architecture of the proposed model-guided deep convolutional network (MoG-DCN) for HSISR. (a) The overall architecture of the proposed network; (b) the architecture of the reconstruction module; (c) the architecture of the degradation operators, and (d) the architecture of the encoding and decoding blocks of the U-net denoiser.

matrix multiplications) and the denoising step (i.e., the non-differential proximal operator) in **Algorithm 1** can be conveniently implemented by the modules of deep convolutional networks.

The overall network architecture of the model-guided deep convolutional network (MoG-DCN) is shown in Fig. 1(a), which executes T iterations of **Algorithm 1**. We have paid special attention to the design of reconstruction and denoising modules - i.e., the former acts as the back-projection operator [10] (step (3) in **Algorithm 1**) and the latter is exactly the proximal operator [26] (step (4) in **Algorithm 1**). Note that although the unfolding strategy has been successfully used for the restoration of natural images [21], [45], [47], similar success has not been witnessed for HSISR yet [10], [13]. When compared with previous approaches [10], [13], the proposed **Algorithm 1** differs significantly in terms of both data and regularization terms, enjoying several advantages such as robustness (due to improved approximation of the observation model) and transparency (thanks to our model-guided design). As shown in Sec. V, the proposed MoG-DCN performs much better than the existing methods [10], [13].

A. The Reconstruction Module

The design of the reconstruction module corresponds to the computing of the intermediate result $\boldsymbol{V}^{(t)}$ as described in Eq. (7) (differential term). As shown in Fig. 1 (a), the initial estimate $\boldsymbol{Z}^{(0)}$ is first updated by the reconstruction module that exactly computes Eq. (7), followed by the DCN denoising module acting as the proximal operator. The architecture of the reconstruction module is shown in Fig. 1 (b), which involves both spatial and spectral degradation operators (i.e., $\boldsymbol{H}$ and $\boldsymbol{P}$) and their transposed versions (i.e., $\boldsymbol{H}^\top$ and $\boldsymbol{P}^\top$). It is worth noting that the symmetric architecture of the reconstruction module reflects the dual role played by spatial and spectral LR images (i.e., top and bottom channels). Conceptually similar to [10], the updating of intermediate result $\boldsymbol{V}^{(t)}$ is based on the fusion of back-projection errors from two parallel channels.

More specifically, we propose to implement the spatial operator $\boldsymbol{H}$ by a convolution of the HSIs with filters along the spatial axis followed by the subsampling operation. To better characterize different spatial blurring matrices (i.e., point spread functions), we opt to approximate $\boldsymbol{H}$ by a devoted subnetwork containing three or four convolutional layers with stride 2 for an overall scaling factor of 8 or 16. Similarly, we approximate $\boldsymbol{H}^\top$ with another subnetwork containing three or four deconvolutional layers, which first upsample the input images along the spatial axis with zero padding, followed by convolutional layers. Since the spectral downsampling operator is relatively simpler, we can model $\boldsymbol{P}$ and $\boldsymbol{P}^\top$ with a single convolutional layer. The architectures of those operators are shown in Fig. 1 (c).

When compared with the existing work on deep HSI fusion [10], we note that a key difference lies in the approximation of $\boldsymbol{H}$ and $\boldsymbol{H}^\top$ by multilayer networks instead of a single-layer one as used in [10]. Such improvement is particularly effective for handling real-world situations when different degradation matrices might be involved with HSI. Our experimental results have shown that more convolutional layers can better approximate multiple degradation matrices than a single convolutional layer in the scenario of blind HSISR (please refer to Sec. V-E). Moreover, unlike [10] based on iterative back-projection, our PGD-based optimization is computationally more efficient, requiring less number of unfolding stages.

B. The DCN Denoising Module

The design of DCN denoising module corresponds to the computing of updated estimate $\boldsymbol{Z}^{(t+1)}$ as described in Eq. (8)

(the non-differential term). As shown in Fig. 1 (a), the intermediate estimate $\boldsymbol{V}^{(t)}$ obtained by the reconstruction module is then fed into the proximal operator (i.e., the denoising module $g(\boldsymbol{Z})$) for further refinement. In this work, for simplicity, we have adopted the U-net [48] as the backbone network architecture for HSI denoising. Other more effective networks for HSI denoising can also be adopted. The U-net denoising network consists of an encoder and a decoder. The encoder includes four encoding blocks (EB), as shown in Fig. 1(d) each of which contains four convolutional layers with 3×3 kernels and ReLU nonlinearity. For the first three EBs, the stride of the last convolutional layer is set to 2 to reduce the spatial dimensions of the feature maps. The decoder also consists of four decoding blocks (DBs), each of which has four convolutional layers with 3×3 kernels and ReLU nonlinearity, as shown in Fig. 1 (d). Except the last DB, each DB is followed by a deconvolution layer to upsample the feature map with scaling factor 2. To compensate the loss of spatial resolution, the upsampled feature maps are then concatenated with the feature maps from the encoder of the same resolution. Instead of predicting the original signal, we let the denoising module to predict the residual by adding a skip connection from the input to the output.

To reduce the number of network parameters and the effect of overfitting, we opt to enforce all T DCN denoising modules sharing the same network parameters. Due to the powerful learning capability of DCN, the DCN denoisers have shown excellent blind denoising performance [44]- i.e., single DCN denoisers can well restore noisy images of different noise levels. We have experimentally found that parameters sharing of T DCN denoising modules does not decrease the reconstruction performance, as shown in Sec. V-B. The unfolding of **Algorithm 1** leads to an overall network architecture containing T denoising modules interleaved with the reconstruction modules. To alleviate the vanishing gradient problem, we propose to connect those feature maps from the previous denoising module to the subsequent denoising modules. As shown in Fig. 1 (a), the feature maps of the EB and DB of the first denoising module are connected to the EBs and DBs of the subsequent denoising modules, respectively. The fusion of those feature maps are achieved by first concatenating them followed by a 1×1 convolutional layer to reduce the number of feature maps. As shown in our ablation studies (please refer to Sec. V-A), the skip connections between the denoising modules significantly improve the HSISR performance.

When compared with those [11], [12] using general "black box" like deep networks for HSISR, our MoG-DCN can fully exploit the observation models of Eq. (1) and the deep prior of HSI learned from training HSIs, which can lead to significant HSISR performance improvements. When compared with previous work such as Deep Blind Iterative Fusion Network (DBIN) [10], our design of the denoising module differs in the following aspects. First, the guiding model behind the DBIN design is the iterative back-projection; while our starting point as described in Eq. (2) is capable of characterizing a wider range of models by varying the regularization term $J(\boldsymbol{Z})$. Accordingly, our unfolding based on the PGD-based optimization can be viewed as the generalization of previous work [10]. Second, in terms of network architecture, (DBIN) [10] used ResNet to denoise the residual signals of HSI; while ours is based on an encoder-decoder configuration directly working with the HSI, which arguably has better capability of restoring HSI more accurately. Additionally, we have also taken both long and short skip connections into account (like ResNet) to expedite the information flow both within and across different stages.

C. Network Training

The overall network is trained by minimizing the following loss function

$$\boldsymbol{\Theta} = \underset{\boldsymbol{\Theta}}{\operatorname{argmin}} \sum_{i=1}^{N} ||\mathcal{F}(\boldsymbol{X}_i, \boldsymbol{Y}_i; \boldsymbol{\Theta}) - \boldsymbol{Z}_i||_1, \tag{9}$$

where $\boldsymbol{X}_i$, $\boldsymbol{Y}_i$ and $\boldsymbol{Z}_i$ denote the i^{th} pair of LR HSI, HR MSI, and the original HR HSI, respectively, and $\mathcal{F}(\cdot, \boldsymbol{\Theta})$ denotes the reconstructed HSI patch by the network with parameters $\boldsymbol{\Theta}$. Our experimental results show that the ℓ_1 loss function performs better than the ℓ_2 loss. We used the ADAM optimizer [49] to train the network with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, and learning rate 10^{-4}. The parameters of the convolutional filters were initialized by the Xavier initializers proposed in [50].

V. Experimental Results

A. Experimental Setup

We have verified the performance of the proposed model-guided deep convolutional networks (MoG-DCN) on two widely used HSI datasets - i.e., the CAVE and Harvard datasets [51]. The CAVE dataset contains 32 512×512 HSIs of indoor scenes, which were captured with 31 spectral bands ranging from 400 to 700 nm at an interval of 10nm. The Harvard dataset contains 50 HSIs of real-world indoor and outdoor scenes, which have 31 spectral bands ranging from 420nm to 720nm. For the CAVE dataset, we have used the first 20 HSIs for training, and the last 12 HSIs for testing; for the Harvard dataset, we have used the first 30 HSIs for training, and the last 20 HSIs for testing. Similar grouping strategy of training and testing datasets has also been adopted in [10].

To generate the training samples, we have used the downsampling matrixes $\boldsymbol{H}$ to simulate the pairs of LR-HSI and HR-MSI, which can be implemented by first applying an anti-aliasing 8×8 Gaussian filter with a standard deviation of two to the original HR-HSIs followed by a downsampling along both the horizontal and vertical dimensions with the scaling factor of eight. Like [10], [14], HR-MSI (i.e., RGB) images were generated by applying the spectral downsampling matrix $\boldsymbol{P}$ constructed from the response of a Nikon D700 camera.[1] To train the proposed network, image patches of size $12 \times 12 \times 31$ and $96 \times 96 \times 3$ were extracted from the LR-HSIs $\boldsymbol{X}_i$ and HR-RGB images $\boldsymbol{Y}_i$, respectively. To generate more training samples, standard data argumentation techniques such

[1] Available at: https://www.maxmax.com/spectral_response.htm

TABLE I

AVERAGE QUANTITATIVE RESULTS OF THE VARIANTS OF THE PROPOSED METHOD WITH DIFFERENT NUMBER OF STAGES ON CAVE AND HARVARD TEST DATASET

Dataset	CAVE				Harvard			
No. of stages	2	3	4	5	2	3	4	5
PSNR(dB)	49.69	49.86	49.89	49.90	47.53	47.57	47.64	47.60
SAM	2.06	2.00	2.04	2.02	2.69	2.69	2.67	2.69
ERGAS	0.46	0.45	0.45	0.453	0.92	0.92	0.91	0.92
SSIM	0.996	0.996	0.996	0.996	0.988	0.988	0.988	0.988

as geometric flipping and rotation were applied to the training images. The batch size was set to 4. We have trained the proposed network with one million iterations. The parameters of the convolutional filters were initialized by the Xavier initializer [50]. The proposed network was implemented with PyTorch platform and trained using 4 Nvidia Titan XP GPUs. To evaluate the performance of the proposed HRHSI methods, four quality metrics, i.e., the peak-signal-to-noise ratio (PSNR), the structural similarity index (SSIM) [52], relative dimensionless global error in synthesis (ERGAS) [53], and spectral angle mapper (SAM) [54] were used in our benchmark study. Note that unlike PSNR and SSIM (the larger the better), ERGAS and SAM are inversely proportional to the image quality (the smaller the better).

B. Ablation Studies

In this subsection, we first verify the effect of the number of stages T and the dense connections between the denoisers on the performance of the proposed MoG-DCN. To support our ablation studies, we have implemented several variants of the proposed network containing different stages. Table I shows the average quantitative results on the CAVE and Harvard datasets by the variants of the proposed networks with different number of stages. It can be seen from Table I that more stages lead to better reconstruction quality and the performance improvement quickly becomes saturated when $T > 3$. Considering the trade-off between performance and complexity, we have set $T = 4$ in our implementation.

We have also implemented several variants of the proposed method including without dense connections, without parameter sharing, and without reconstruction module. For variant of the proposed method without using the reconstruction module, the initially interpolated HSI $\boldsymbol{Z}^{(0)}$ concatenated with the HR-MSI image $\boldsymbol{Y}$ was fed into the DCN denoiser, whose output concatenated with $\boldsymbol{Y}$ was fed into the next DCN denoiser. The comparison study results on CAVE and Harvard datasets are shown in Table II. From Table II, we can see that all three components (dense connections, parameter sharing, and reconstruction module) further improve the HSI reconstruction performance for both CAVE and Harvard datasets. The reduction of ERGAS and SAM are more noticeable than PSNR/SSIM improvement in the presence of dense connections and reconstruction module. Such experimental findings have confirmed the benefit of those components to facilitate the information flow during the training of MoG-DCN model. To verify the effectiveness of U-net denoising modules, we have also implemented a variant of the proposed method where U-net denoising modules were replaced by ResNet denoising modules adopted in [10] containing a similar number of parameters. From Table III, we can see that the proposed method with U-net denoisers outperformed that of ResNet denoisers by up to 1.2 dB on the CAVE dataset.

TABLE II

ABLATION STUDY ON THE EFFECTS OF DENSE CONNECTIONS, PARAMETER SHARING, AND RECONSTRUCTION MODULE FOR CAVE AND HARVARD DATASETS

Dataset	CAVE		Harvard	
Dense connections	✔	✗	✔	✗
PSNR(dB)	49.89	49.60	47.60	47.341
ERGAS	0.45	0.47	0.91	0.95
SAM	2.04	2.04	2.68	2.71
SSIM	0.996	0.996	0.988	0.988
Sharing params	✔	✗	✔	✗
PSNR(dB)	49.89	49.85	47.60	47.58
SAM	2.04	2.00	2.68	2.69
ERGAS	0.45	0.45	0.91	0.91
SSIM	0.996	0.996	0.988	0.988
with Recon	✔	✗	✔	✗
PSNR(dB)	49.89	49.40	47.60	47.55
SAM	2.04	2.15	2.68	2.71
ERGAS	0.45	0.47	0.91	0.92
SSIM	0.996	0.996	0.988	0.988

TABLE III

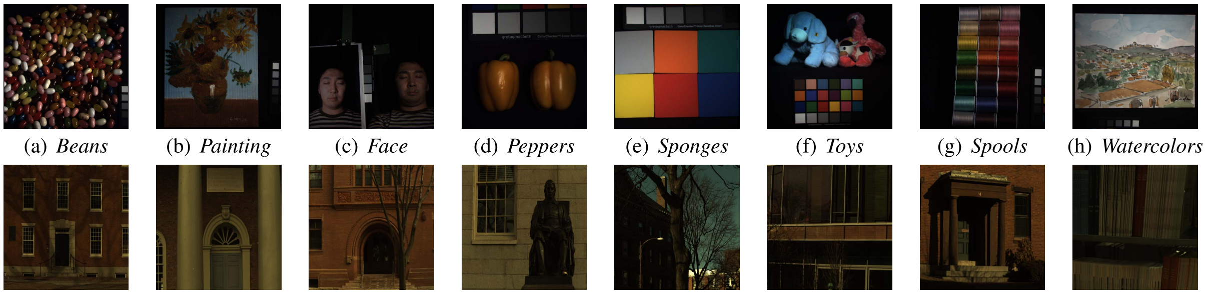
THE AVERAGE RESULTS OF THE PROPOSED METHODS USING THE U-NET AND RESNET DENOISERS ON CAVE AND HARVARD DATASETS FOR SCALING FACTOR 8

Dataset	CAVE		Harvard	
Denoiser	U-net	ResNet	U-net	ResNet
PSNR(dB)	49.89	48.69	47.60	47.32
SAM	2.04	2.27	2.68	2.74
ERGAS	0.45	0.51	0.91	0.95
SSIM	0.996	0.995	0.988	0.988

C. Comparison Study on CAVE and Harvard Datasets

We have compared the proposed method with several state-of-the-art HSISR methods, including the coupled sparse tensor factorization method (CSTF) [55], the nonnegative structured sparse representation method (NSSR) [14], the subspace regularized method (HySure) [56], the Bayesian sparse representation method (BSR) [36], the tensor low-rank based method (LTTR) [38], and the three recently proposed deep learning based methods - i.e., the MSI and HSI fusion net (MHF-net) [13], the deep blind iterative fusion network (DBIN) [10], and the CNN denoising based method (CNN-FUS) [57]. For the reason of fairness, all deep neural network-based methods including the proposed method have been retrained on the same training dataset. The results of conventional HSISR methods were generated with the original source codes downloaded from the authors' website. The average PSNR, SSIM, ERGAS, and SAM results on the two datasets are shown in Tables IV and V.

From Table IV, we can see that the recently proposed deep learning based methods (e.g., MHF-net [13] and DBIN [10]) outperform other conventional HSISR methods by a large margin. When compared with MHF-net [13] and DBIN [10]



(a) *Beans* (b) *Painting* (c) *Face* (d) *Peppers* (e) *Sponges* (f) *Toys* (g) *Spools* (h) *Watercolors*
(i) *Image 1* (j) *Image 2* (k) *Image 3* (l) *Image 4* (m) *Image 5* (n) *Image 6* (o) *Image 7* (p) *Image 8*

Fig. 2. The HR RGB images from the CAVE (the first row) and Harvard (the second row) [51] datasets.

TABLE IV

THE AVERAGE PSNR, SAM, ERGAS, AND SSIM RESULTS OF THE TEST METHODS ON THE CAVE DATASET FOR GAUSSIAN BLUR KERNEL AND SCALING FACTORS 8 AND 16 (BEST RESULTS ARE HIGHLIGHTED IN BOLD)

Methods	Hysure [56]	CSTF [55]	NSSR [14]	BSR [36]	DBIN [10]	MHF-net [13]	LTTR [38]	CNN-FUS [57]	**MoG-DCN**
					s = 8				
PSNR(dB)	40.06	42.41	44.07	41.49	48.82	46.31	45.89	44.21	**49.89**
SAM	9.66	5.04	4.35	5.85	2.09	3.39	2.97	4.04	**2.04**
ERGAS	1.30	0.87	0.82	1.10	0.50	0.64	0.66	0.82	**0.45**
SSIM	0.976	0.979	0.987	0.983	0.996	0.994	0.993	0.989	**0.996**
					s = 16				
PSNR(dB)	36.74	41.17	38.97	40.83	43.70	44.51	42.48	40.37	**46.84**
SAM	15.05	6.17	6.73	6.38	3.00	4.00	4.25	5.85	**2.62**
ERGAS	0.88	0.54	1.46	0.59	0.46	0.38	0.47	0.59	**0.31**
SSIM	0.965	0.976	0.977	0.981	0.994	0.992	0.987	0.979	**0.995**

TABLE V

THE AVERAGE PSNR, SAM, ERGAS, AND SSIM RESULTS OF THE TEST METHODS ON THE HARVARD DATASET FOR GAUSSIAN BLUR KERNEL AND SCALING FACTORS 8 AND 16 (BEST RESULTS ARE HIGHLIGHTED IN BOLD)

Methods	Hysure [56]	CSTF [55]	NSSR [14]	BSR [36]	DBIN [10]	MHF-net [13]	LTTR [38]	CNN-FUS [57]	**MoG-DCN**
					s=8				
PSNR(dB)	44.26	44.98	46.08	46.30	47.36	46.42	46.86	46.05	**47.64**
SAM	3.75	3.54	3.40	3.00	2.71	3.01	2.90	3.24	**2.67**
ERGAS	1.40	1.07	1.20	1.11	0.97	1.09	1.11	1.12	**0.91**
SSIM	0.983	0.980	0.985	0.986	0.988	0.987	0.987	0.985	**0.988**
					s=16				
PSNR(dB)	42.77	45.17	44.23	45.75	46.27	46.23	45.82	43.47	**46.43**
SAM	4.54	3.76	3.91	3.16	2.94	3.09	3.11	5.41	**2.93**
ERGAS	0.78	0.57	1.53	0.58	0.53	0.54	0.65	0.92	**0.53**
SSIM	0.981	0.983	0.983	0.986	0.987	0.987	0.986	0.966	**0.987**

methods, the proposed method performs better than both. The average PSNR gain over the DBIN method (the second best) is 0.86 dB for $\times 8$ case. For $\times 16$ case, the proposed method outperforms the MHF-net method [13] (the second best) by 2.33 dB on average. Similar observations can also be made on the Harvard dataset as shown in Table V. Note that since the Harvard dataset is less challenging than the CAVE dataset, conventional model-based methods can also achieve good HSISR performance. However, deep learning based methods still outperform all model-based ones. For this dataset, the proposed method outperforms DBIN [10] (the second best) by up to 0.28 dB and 0.16 dB on the average for $\times 8$ and $\times 16$, respectively. Parts of the reconstructed HR HSIs by different competing methods are shown in Figs. 3-6. From Figs. 3-6, one can see that the proposed method can recover more details of HR-HSIs than other competing methods.

D. Experiments on Real Multispectral Images

We have also verified the performance of the proposed method on a real-world multispectral image (MSI) dataset, called WV2.[2] The dataset contains a pair of real LR MSI containing eight bands of spatial size 418×658 and a HR RGB image of size $1676 \times 2632 \times 3$. We aim to reconstruct a HR MSI from the pair of LR MSI and the HR RGB image. Since the datasets only contain one pair of images, similar to [10], [13] we used the top half of the HR RGB and the LR MSI to train the proposed MoG-DCN, while using the remaining part of the dataset for testing. As the ground truth of the HR MSI is unavailable, we have used the Wald's protocol [58] to generate the training samples. A total 18000 pairs of RGB image blocks of size $64 \times 64 \times 3$ and LR MSI block of size $16 \times 16 \times 8$ were generated. We compared the proposed method with the HySure method [56] and the two recently proposed deep learning-based methods - i.e., the MHF-net method [13] and DBIN method [10]. Parts of the reconstructed HR MSIs were shown in Fig. 7. While all deep learning based methods clearly outperform the conventional HySure

[2] Available at: https://www.harrisgeospatial.com/Data-Imagery/Satellite-Imagery/High-Resolution/WorldView-2

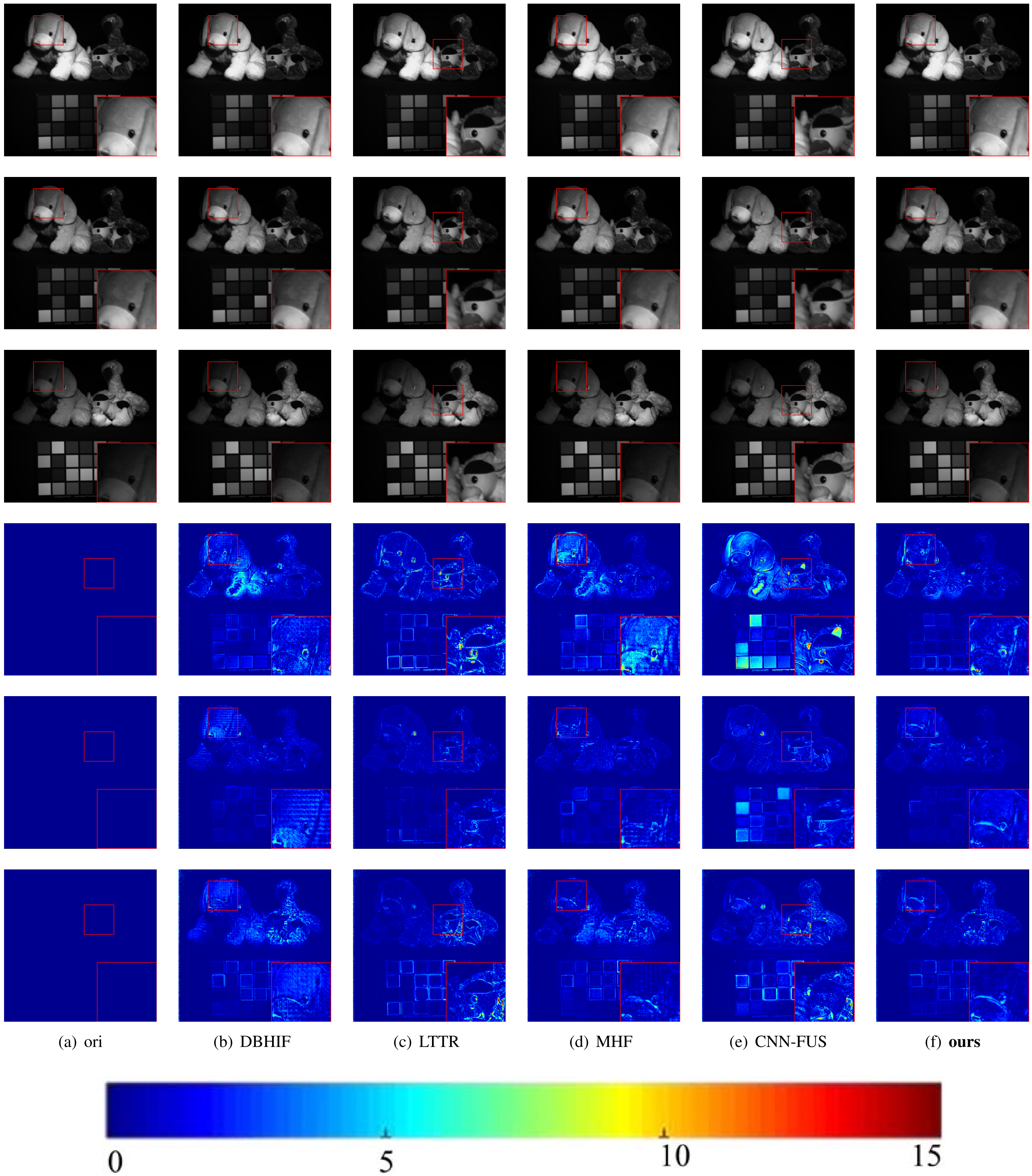


Fig. 3. Reconstructed images of *stuffed_toys* in CAVE dataset at 480nm, 560nm and 670nm with gaussian blur kernel and scaling factor $s = 8$. The first 3 rows show the reconstructed images for 480nm, 560nm and 670nm bands, respectively; the last 3 rows show the error images of the competing methods. (a) the ground_truth images; (b) the DBHIF method (PSNR = 46.65 dB, ERGAS = 0.49, SAM = 2.60, SSIM = 0.996); (c) the LTTR method (PSNR = 46.524, ERGAS = 0.49, SAM = 3.04,SSIM = 0.995); (d) the MHF method (PSNR = 44.94 dB, ERGAS = 0.583, SAM = 4.25, SSIM = 0.994); (e) the CNN-FUS method (PSNR = 44.79 dB, ERGAS = 0.61, SAM = 4.73, SSIM = 0.992); (f) the **proposed MoG-DCN** method (PSNR = **47.41** dB, ERGAS = **0.44**, SAM = **2.60s**, SSIM = **0.996**).

method [56], the proposed method performs better than the MHF-net [13] and DBIN [10] in suppressing visual artifacts.

E. Blind HSI Super-Resolution

In the above experimental comparison studies, the degradation matrices $\boldsymbol{H}$ and $\boldsymbol{P}$ are assumed to be known for the conventional HSISR methods, or explicitly assumed to be known by generating the pairs of LR-HSI and HR-MSI images by the given $\boldsymbol{H}$ and $\boldsymbol{P}$ during the training of deep models. However, such a nonblind assumption about the known degradation matrices $\boldsymbol{H}$ and $\boldsymbol{P}$ is often unrealistic for real-world applications. To demonstrate the potential of the proposed MoG-DCN for real-world HSISR applications, we investigate the blind HSISR problem without the assumption about the

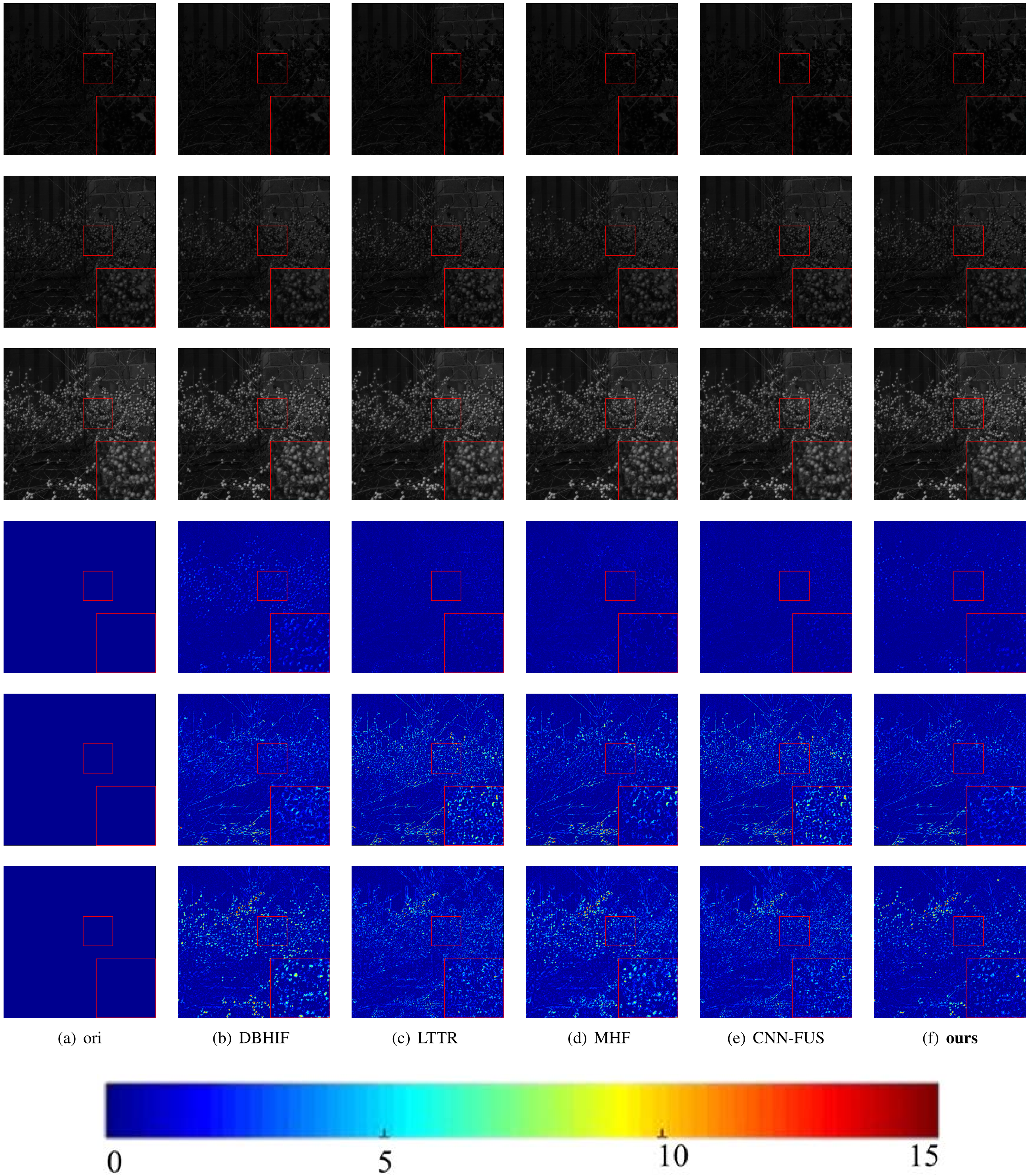


Fig. 4. Reconstructed images of *imgf8* in HARVARD dataset at 540nm, 590nm and 680nm with gaussian blur kernel and scaling factor $s = 8$. The first 3 rows show the reconstructed images for 540nm, 590nm and 680nm bands, respectively; the last 3 rows show the error images of the competing methods. (a) the ground_truth images; (b) the DBHIF method (PSNR = 47.89 dB, ERGAS = 1.21, SAM = 3.18, SSIM = 0.994); (c) the LTTR method (PSNR = 47.44, ERGAS = 1.56, SAM = 3.46, SSIM = 0.993); (d) the MHF method (PSNR = 47.31dB, ERGAS = 1.41, SAM = 3.47, SSIM = 0.993); (e) the CNN-FUS method (PSNR = 47.24 dB, ERGAS = 1.45, SAM = 3.54, SSIM = 0.993); (f) the **proposed MoG-DCN** method (PSNR = **48.69** dB, ERGAS = **1.12**, SAM = **3.02**, SSIM = **0.995**).

known $\boldsymbol{H}$. As the spectral transform matrix $\boldsymbol{P}$ is often given, it is plausible to assume $\boldsymbol{P}$ to be known for simplicity. In the last study of blind HSISR, we assume the blur kernel for generating the LR-HSIs is a Gaussian kernel of 8×8 with the standard deviation sampled from the range of [1.0, 3.0], which is realistic for practical applications. The challenge with blind HSISR is to reconstruct HR-HSI images without the exact knowledge about the blur kernel.

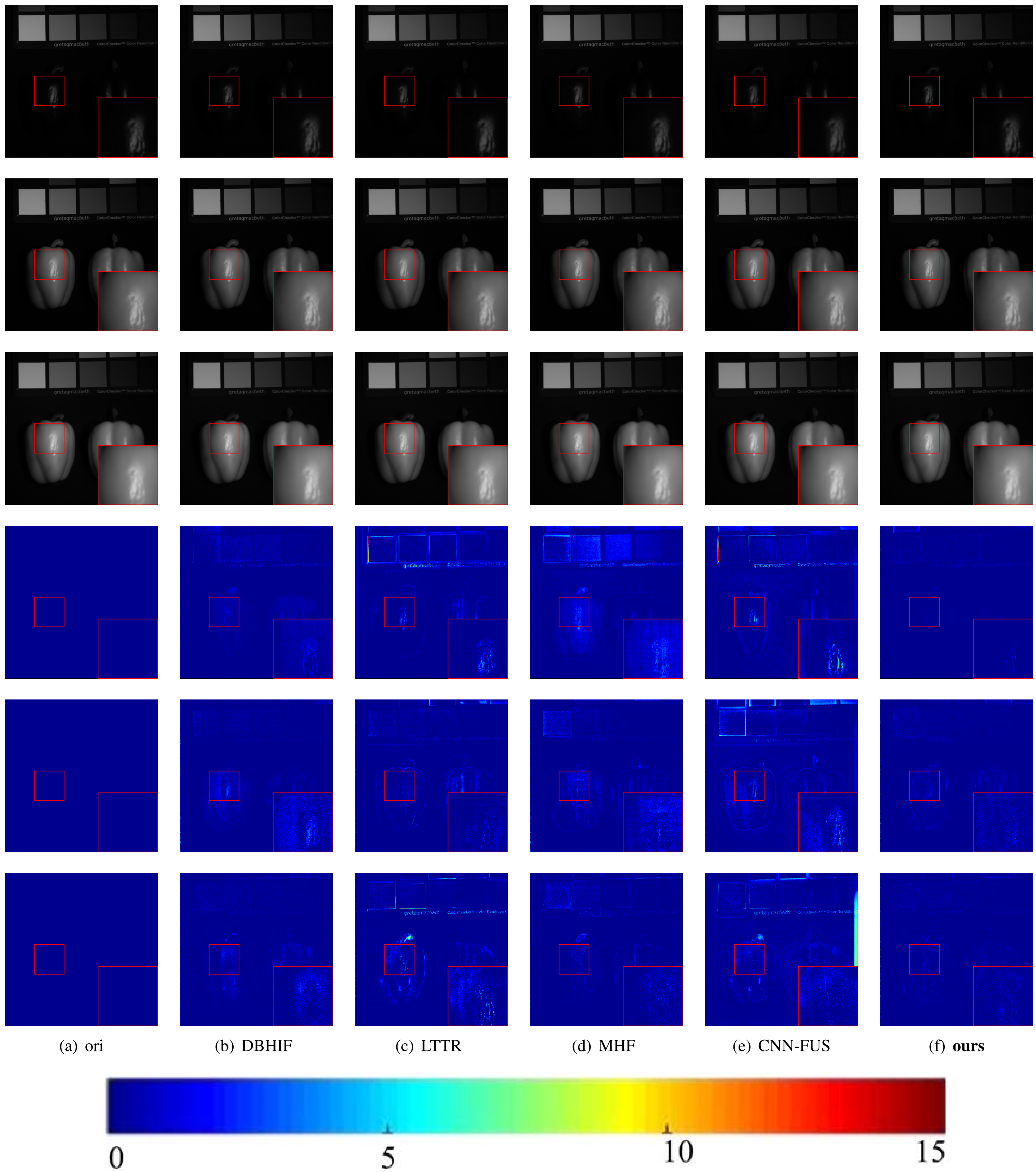


Fig. 5. Reconstructed images of *real_and_fake_peppers* in CAVE dataset at 470nm, 570nm and 680nm with gaussian blur kernel and scaling factor $s = 16$. The first 3 rows show the reconstructed images for 470nm, 570nm and 680nm bands, respectively; the last 3 rows show the error images of the competing methods. (a) the ground_truth images; (b) the DBHIF method (PSNR = 55.19 dB, ERGAS = 0.14, SAM = 1.50, SSIM = 0.999); (c) the LTTR method (PSNR = 47.50, ERGAS = 0.67, SAM = 2.78, SSIM = 0.996); (d) the MHF method (PSNR = 50.59 dB, ERGAS = 0.24, SAM = 2.56, SSIM = 0.997); (e) the CNN-FUS method (PSNR = 45.46 dB, ERGAS = 0.43, SAM = 4.35, SSIM = 0.988); (f) the **proposed MoG-DCN** method (PSNR = **55.78** dB, ERGAS = **0.13**, SAM = **1.49**, SSIM = **0.999**).

To deal with the uncertainty in the parameters of blur kernels, we propose to simulate the LR-HSIs with different Gaussian blur kernels randomly sampled from the range of $[1.0, 3.0]$ and train the proposed MoG-DCN model on this training sample set to deal with multiple blurring kernels. Since the degradation matrix $\boldsymbol{H}$ is approximated with a sub-CNN in the proposed method, the proposed method can be expected to well super-resolve LR-HSIs with unknown

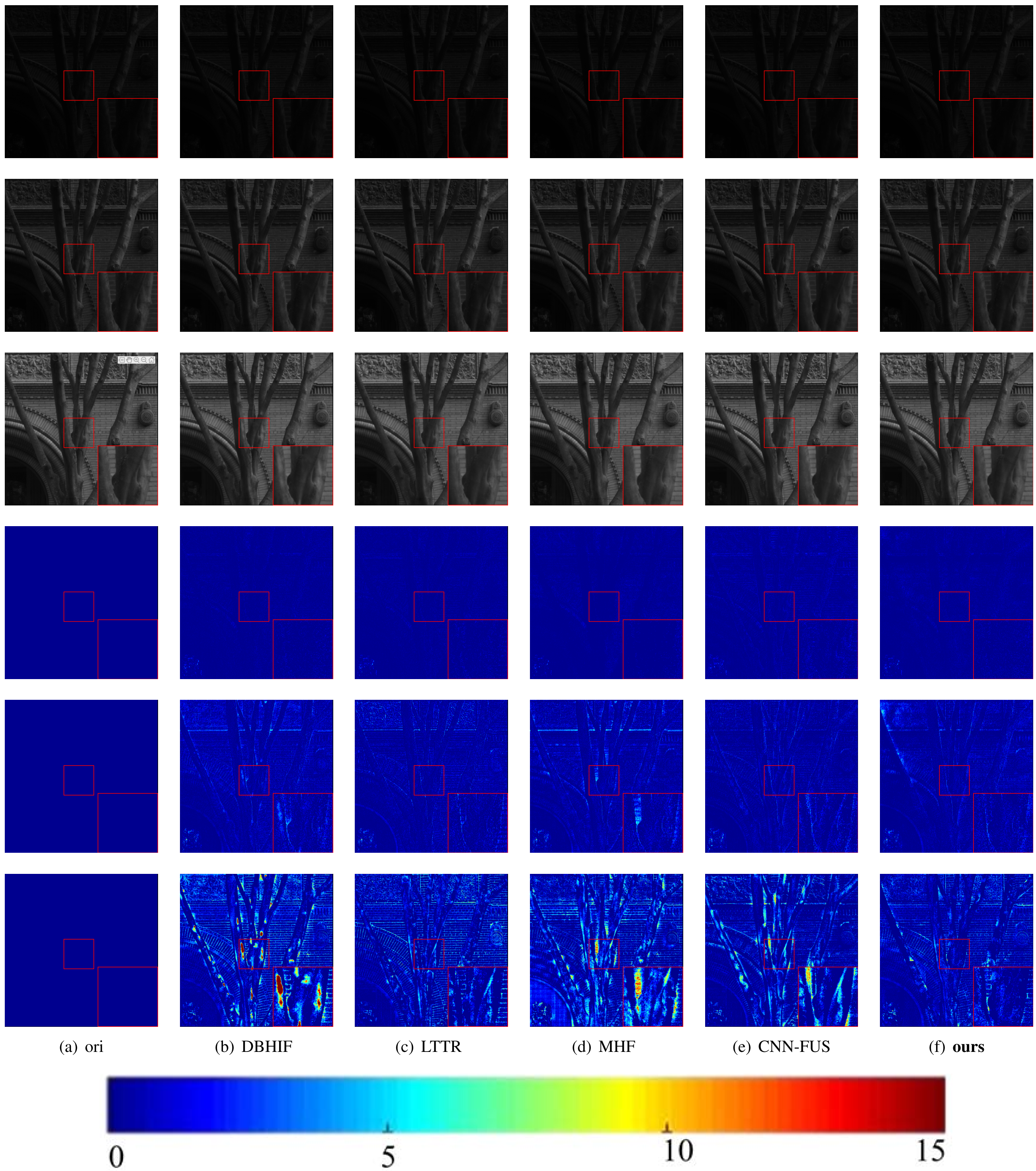
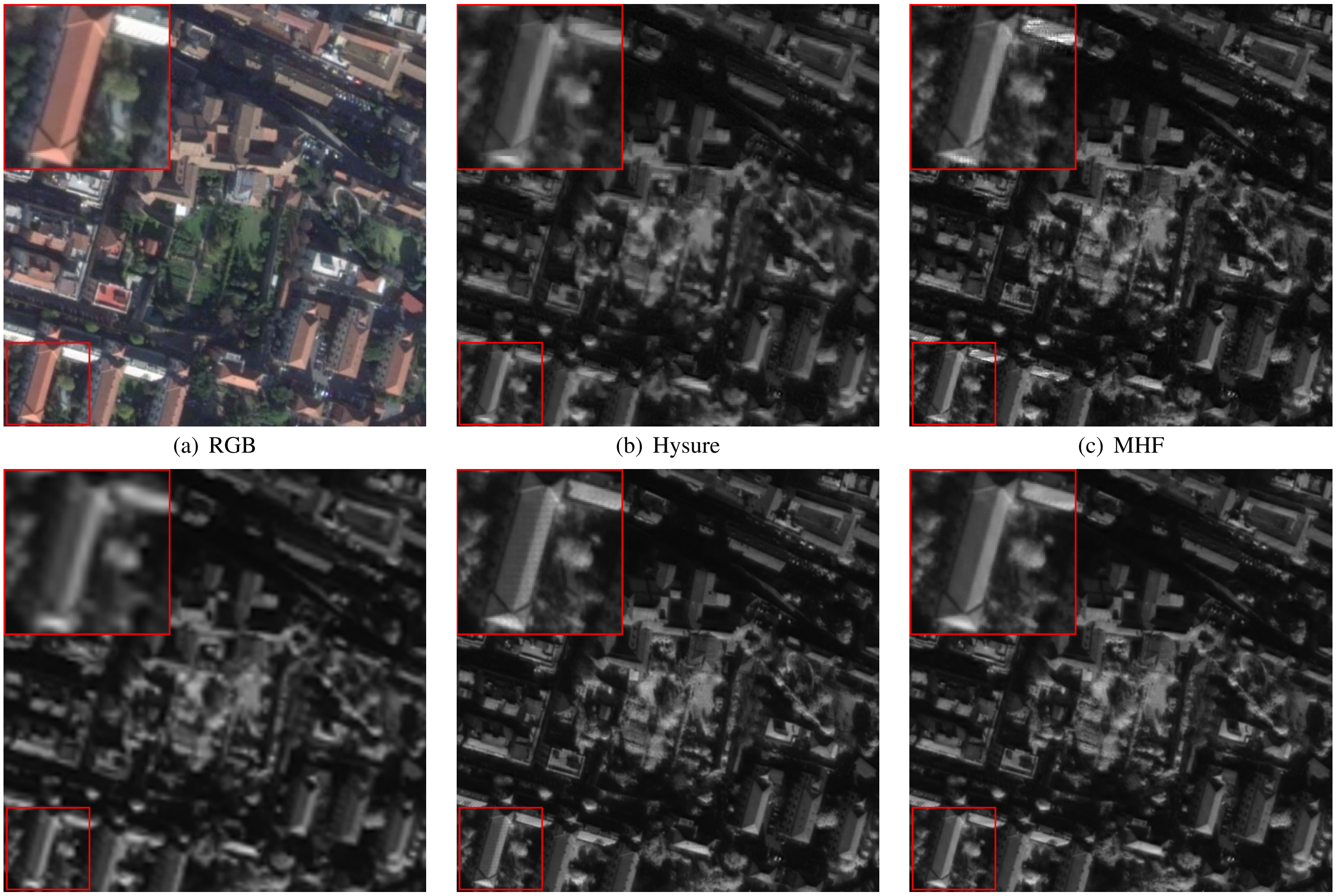


Fig. 6. Reconstructed images of *image6* in HARVARD dataset at 450nm, 530nm and 700nm with gaussian blur kernel and scaling factor $s = 16$. The first 3 rows show the reconstructed images for 450nm, 530nm and 700nm bands, respectively; the last 3 rows show the error images of the competing methods. (a) the ground_truth images; (b) the DBHIF method (PSNR = 45.91 dB, ERGAS = 0.30, SAM = 1.72, SSIM = 0.991); (c) the LTTR method (PSNR = 44.42, GRGAS = 0.86, SAM = 1.99, SSIM = 0.988); (d) the MHF method (PSNR = 45.74 dB, ERGAS = 0.31, SAM = 1.84, SSIM = 0.991); (e) the CNN-FUS method (PSNR = 45.32 dB, ERGAS = 0.394, SAM = 1.97, SSIM = 0.989); (f) the **proposed MoG-DCN** method (PSNR = **46.13** dB, ERGAS = **0.29**, SAM = **1.71**, SSIM = **0.991**).

subsampling matrices $\boldsymbol{H}$. To show the effectiveness of the proposed method for blind HSISR, we have compared with the MHF-net method [13] and the DBIN method [10], both of which were also trained on the same training sample set for the reason of fairness. Table VI shows the average results on the CAVE dataset by the competing methods for scaling factor 8 and different Gaussian blur kernels. From table VI, we can see that all three deep learning based methods can



(a) RGB (b) Hysure (c) MHF

(d) LR-MSI (e) DBHIF (f) Ours

Fig. 7. Reconstructed images of World View-2.

TABLE VI

THE AVERAGE BLIND HSISR RESULTS OF THE TEST METHODS ON THE CAVE DATASET WITH GAUSSIAN BLUR KERNELS WITH DIFFERENT STANDARD DEVIATIONS σ FOR A SCALING FACTOR OF 8

σ	1.0	1.2	1.4	1.6	1.8	2.0	2.2	2.4	2.6	2.8	3.0
MoG-DCN (ours)											
PSNR	49.37	49.55	49.67	49.75	49.79	49.82	49.83	49.84	49.84	49.83	49.83
ERGAS	0.48	0.47	0.47	0.46	0.46	0.46	0.46	0.46	0.46	0.46	0.46
SAM	2.09	2.06	2.04	2.04	2.03	2.03	2.03	2.03	2.04	2.04	2.04
SSIM	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996
DBIN [10]											
PSNR	47.58	47.66	47.71	47.75	47.76	47.77	47.78	47.78	47.78	47.77	47.77
ERGAS	0.56	0.55	0.55	0.55	0.54	0.54	0.54	0.54	0.54	0.54	0.54
SAM	2.30	2.29	2.28	2.27	2.27	2.26	2.26	2.27	2.27	2.27	2.27
SSIM	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996
MHF-net [13]											
PSNR	46.18	46.29	46.40	46.42	46.45	46.47	46.48	46.49	46.49	46.49	46.49
ERGAS	0.65	0.65	0.64	0.64	0.63	0.63	0.63	0.63	0.63	0.63	0.63
SAM	3.39	3.36	3.34	3.33	3.32	3.32	3.32	3.32	3.32	3.32	3.32
SSIM	0.994	0.994	0.994	0.994	0.994	0.994	0.994	0.994	0.994	0.994	0.994

well reconstruct the HR-HSIs for different blur kernels. The proposed method outperforms the MHF-net method [13] and the DBIN method [10] for all blur kernels, verifying the superior performance of the proposed method for blind HSISR.

Considering that the blur kernel would become more complicated in real-world scenarios, we have also conducted experiments to verify the blind HSI reconstruction performance of the proposed method when there were mismatches of blur kernels between the model training and testing. To simulate such mismatches, both the Gaussian blur kernels with deviations out of the range (i.e., [1.0, 3.0]) and the bicubic kernel were used to generate the LR HSIs. As shown in Table VII,

TABLE VII

THE AVERAGE BLIND HSISR RESULTS OF THE TEST METHODS ON THE CAVE DATASET FOR SCALING FACTOR OF 8 WITH GAUSSIAN KERNELS WITH σ OUT OF RANGE [1.0, 3.0] AND BICUBIC KERNEL (GAUSSIAN KERNELS WITH σ IN THE RANGE [1.0, 3.0] WERE USED FOR TRAINING)

σ	0.4	0.6	0.8	3.2	3.4	3.6	Bic.
MoG-DCN							
PSNR(dB)	48.79	48.93	49.23	49.81	49.81	49.80	47.18
SAM	2.216	2.185	2.125	2.057	2.060	2.062	2.674
ERGAS	0.509	0.502	0.486	0.455	0.455	0.455	0.598
SSIM	0.996	0.996	0.996	0.996	0.996	0.996	0.995
DBIN [10]							
PSNR(dB)	47.25	47.31	47.45	47.77	47.76	47.76	46.97
SAM	2.371	2.355	2.318	2.271	2.273	2.275	2.501
ERGAS	0.577	0.573	0.565	0.543	0.543	0.543	0.595
SSIM	0.995	0.995	0.995	0.995	0.995	0.995	0.995
MHF-net [13]							
PSNR(dB)	45.69	45.76	45.92	46.39	46.39	46.39	45.58
SAM	3.346	3.330	3.297	3.268	3.270	3.273	3.525
ERGAS	0.681	0.676	0.665	0.632	0.632	0.633	0.696
SSIM	0.994	0.994	0.994	0.994	0.994	0.994	0.993

the performance of the proposed method decreased when there were blur kernel mismatches, especially for the class of bicubic kernels different from the assumed Gaussian blur kernels. Compared with other competing methods, the proposed method still outperformed other methods when there were mismatches of blur kernels between model training and testing.

VI. CONCLUSION

In this paper, we have presented a new PGD-based iterative HSISR algorithm and shown how to unfold it by deep convolutional network implementation. Our MoG-DCN aims at better exploiting HSI-related domain knowledge and optimizing DCN-based HSISR algorithm in a principled (model-guided) manner. By taking the observation matrix of HSI into the consideration, we show how to unfold an iterative HSISR algorithm into a novel model-guided deep convolutional network (MoG-DCN). In addition to during end-to-end optimization, the representation of the observation matrix by subnetworks also allows the unfolded deep HSISR network to work with different HSI situations, which greatly improves the generalization property of the proposed MoG-DCN. Our experimental results have shown that the proposed MoG-DCN outperforms several current leading HSISR methods by striking a better trade-off between the model cost and HSISR performance. Additionally, we have verified the excellent generalization property of MoG-DCN in the situation of blind HSISR, which significantly outperform existing state-of-the-art methods.

Inspired by the latest work on tuning-free Plug-and-Play (PnP) proximal algorithm [59], it is possible to further optimize our MoG-DCN for blind HSISR. By treating internal parameter selection of HSISR as a sequential decision making problem, we can develop a policy network for automatic search of parameters. Such an idea has been shown effective in compressed sensing via mixed model-free and model-based deep reinforcement learning [59]. One possible extension is to leverage this idea to automatically determine the internal parameters of unfolded networks including the penalty parameter, the denoising module, and the number of stages. Another possible extension is to pursue an alternating direction method of multipliers (ADMM) instead of the proximal gradient method. To address the issue of heavy computation and memory requirements, it is possible to consider an incremental variant of the widely used PnP-ADMM algorithm [17], making it scalable to large-scale datasets [60]. We leave these lines of research as our future works.

REFERENCES

[1] D. Manolakis and G. Shaw, "Detection algorithms for hyperspectral imaging applications," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 29–43, Aug. 2002.

[2] Y. Tarabalka, J. Chanussot, and J. A. Benediktsson, "Segmentation and classification of hyperspectral images using minimum spanning forest grown from automatically selected markers," *IEEE Trans. Syst., Man, Cybern. B, Cybernetics*, vol. 40, no. 5, pp. 1267–1279, Oct. 2010.

[3] M. Uzair, A. Mahmood, and A. Mian, "Hyperspectral face recognition using 3D-DCT and partial least squares," in *Proc. Brit. Mach. Vis. Conf.*, 2013, pp. 1–57.

[4] Q. Wang, J. Lin, and Y. Yuan, "Salient band selection for hyperspectral image classification via manifold ranking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 6, pp. 1279–1289, Jun. 2016.

[5] B. Uzkent, M. J. Hoffman, and A. Vodacek, "Real-time vehicle tracking in aerial video using hyperspectral features," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2016, pp. 36–44.

[6] B. Uzkent, A. Rangnekar, and M. J. Hoffman, "Aerial vehicle tracking by adaptive fusion of hyperspectral likelihood maps," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 39–48.

[7] H. Van Nguyen, A. Banerjee, and R. Chellappa, "Tracking via object reflectance using a hyperspectral video camera," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2010, pp. 44–51.

[8] W. Xie, J. Lei, Y. Cui, Y. Li, and Q. Du, "Hyperspectral pansharpening with deep priors," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 5, pp. 1529–1543, May 2020.

[9] W. Xie, Y. Cui, Y. Li, J. Lei, Q. Du, and J. Li, "HPGAN: Hyperspectral pansharpening using 3-D generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 463–477, Jan. 2021.

[10] W. Wang, W. Zeng, Y. Huang, X. Ding, and J. Paisley, "Deep blind hyperspectral image fusion," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4150–4159.

[11] F. Palsson, J. R. Sveinsson, and M. O. Ulfarsson, "Multispectral and hyperspectral image fusion using a 3-D-Convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 639–643, May 2017.

[12] G. Scarpa, S. Vitale, and D. Cozzolino, "Target-adaptive CNN-based pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5443–5457, Sep. 2018.

[13] Q. Xie, M. Zhou, Q. Zhao, D. Meng, W. Zuo, and Z. Xu, "Multispectral and hyperspectral image fusion by MS/HS fusion net," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1585–1594.

[14] W. Dong *et al.*, "Hyperspectral image super-resolution via non-negative structured sparse representation," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2337–2352, May 2016.

[15] Q. V. Le, J. Ngiam, A. Coates, A. Lahiri, B. Prochnow, and A. Y. Ng, "On optimization methods for deep learning," in *Proc. ICML*, 2011, pp. 265–272.

[16] K. Gregor and Y. LeCun, "Learning fast approximations of sparse coding," in *Proc. 27th Int. Conf. Int. Conf. Mach. Learn.*, 2010, pp. 399–406.

[17] Y. Yang, J. Sun, H. Li, and Z. Xu, "Deep ADMM-Net for compressive sensing MRI," in *Proc. 30th Int. Conf. Neural Inf. Process. Syst.*, 2016, pp. 10–18.

[18] S. Wisdom, T. Powers, J. Pitton, and L. Atlas, "Building recurrent networks by unfolding iterative thresholding for sequential sparse recovery," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Mar. 2017, pp. 4346–4350.

[19] M. Borgerding, P. Schniter, and S. Rangan, "AMP-inspired deep networks for sparse linear inverse problems," *IEEE Trans. Signal Process.*, vol. 65, no. 16, pp. 4293–4308, Aug. 2017.

[20] C. Chen, Z. Xiong, X. Tian, and F. Wu, "Deep boosting for image denoising," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–18.

[21] W. Dong, P. Wang, W. Yin, G. Shi, F. Wu, and X. Lu, "Denoising prior driven deep neural network for image restoration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 10, pp. 2305–2318, Oct. 2019.

[22] E. Tsiligianni and N. Deligiannis, "Deep coupled-representation learning for sparse linear inverse problems with side information," *IEEE Signal Process. Lett.*, vol. 26, no. 12, pp. 1768–1772, Dec. 2019.

[23] C. Bertocchi, E. Chouzenoux, M.-C. Corbineau, J.-C. Pesquet, and M. Prato, "Deep unfolding of a proximal interior point method for image restoration," *Inverse Problems*, vol. 36, no. 3, Mar. 2020, Art. no. 034005.

[24] C. Molnar, *Interpretable Machine Learning*. Abu Dhabi, United Arab Emirates: Lulu, 2020.

[25] K. C. Toh and S. Yun, "An accelerated proximal gradient algorithm for nuclear norm regularized linear least squares problems," *Pacific J. Optim.*, vol. 6, nos. 615–640, p. 15, 2010.

[26] N. Parikh and S. Boyd, "Proximal algorithms," *Found. Trends Optim.*, vol. 1, no. 3, pp. 127–239, Jan. 2014.

[27] J. Nunez, X. Otazu, O. Fors, A. Prades, V. Pala, and R. Arbiol, "Multiresolution-based image fusion with additive wavelet decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 3, pp. 1204–1211, May 1999.

[28] B. Aiazzi, S. Baronti, F. Lotti, and M. Selva, "A comparison between global and context-adaptive pansharpening of multispectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 2, pp. 302–306, Apr. 2009.

[29] R. Zurita-Milla, J. Clevers, and M. E. Schaepman, "Unmixing-based landsat TM and MERIS FR data fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 3, pp. 453–457, Jul. 2008.

[30] X. Cao, H. Du, X. Tong, Q. Dai, and S. Lin, "A prism-mask system for multispectral video acquisition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 12, pp. 2423–2435, Dec. 2011.

[31] R. Kawakami, Y. Matsushita, J. Wright, M. Ben-Ezra, Y.-W. Tai, and K. Ikeuchi, "High-resolution hyperspectral imaging via matrix factorization," in *Proc. CVPR*, Jun. 2011, pp. 2329–2336.

[32] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[33] R. Dian, L. Fang, and S. Li, "Hyperspectral image super-resolution via non-local sparse tensor factorization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5344–5353.

[34] E. Wycoff, T.-H. Chan, K. Jia, W.-K. Ma, and Y. Ma, "A non-negative sparse promoting algorithm for high resolution hyperspectral imaging," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, May 2013, pp. 1409–1413.

[35] R. Dian, S. Li, and L. Fang, "Non-local sparse representation for hyperspectral image super-resolution," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*. Phoenix, AZ, USA: Springer, Sep. 2016, pp. 63–78.

[36] Q. Wei, J. Bioucas-Dias, N. Dobigeon, and J.-Y. Tourneret, "Hyperspectral and multispectral image fusion based on a sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3658–3668, Jul. 2015.

[37] L. Zhang, W. Wei, C. Bai, Y. Gao, and Y. Zhang, "Exploiting clustering manifold structure for hyperspectral imagery super-resolution," *IEEE Trans. Image Process.*, vol. 27, no. 12, pp. 5969–5972, Aug. 2018.

[38] R. Dian, S. Li, and L. Fang, "Learning a low tensor-train rank representation for hyperspectral image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2672–2683, Sep. 2019.

[39] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.

[40] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[41] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[42] L. Liu *et al.*, "Deep learning for generic object detection: A survey," 2018, *arXiv:1809.02165*. [Online]. Available: http://arxiv.org/abs/1809.02165

[43] C. Dong, C. C. Loy, K. He, and X. Tang, "Learning a deep convolutional network for image super-resolution," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2014, pp. 184–199.

[44] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising," *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3142–3155, Jul. 2017.

[45] K. Zhang, W. Zuo, S. Gu, and L. Zhang, "Learning deep CNN denoiser prior for image restoration," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2808–2817.

[46] T. Meinhardt, M. Moeller, C. Hazirbas, and D. Cremers, "Learning proximal operators: Using denoising networks for regularizing inverse imaging problems," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1799–1808.

[47] F. Kokkinos and S. Lefkimmiatis, "Iterative residual CNNs for burst photography applications," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5929–5938.

[48] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.

[49] D. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. ICLR*, 2014.

[50] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1026–1034.

[51] F. Yasuma, T. Mitsunaga, D. Iso, and S. K. Nayar, "Generalized assorted pixel camera: Postcapture control of resolution, dynamic range, and spectrum," *IEEE Trans. Image Process.*, vol. 19, no. 9, pp. 2241–2253, Sep. 2010.

[52] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[53] L. Wald, "Quality of high resolution synthesised images: Is there a simple criterion?" in *Proc. Int. Conf. Fusion Earth Data*, 2000, pp. 99–103.

[54] Y. Zhang, S. De Backer, and P. Scheunders, "Noise-resistant wavelet-based Bayesian fusion of multispectral and hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 11, pp. 3834–3843, Nov. 2009.

[55] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.

[56] M. Simoes, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[57] R. Dian, S. Li, and X. Kang, "Regularizing hyperspectral and multispectral image fusion by CNN denoiser," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 3, pp. 1124–1135, Mar. 2021.

[58] R. Carla, L. Santurri, B. Aiazzi, and S. Baronti, "Full-scale assessment of pansharpening through polynomial fitting of multiscale measurements," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 12, pp. 6344–6355, Dec. 2015.

[59] K. Wei, A. Aviles-Rivero, J. Liang, Y. Fu, C.-B. Schönlieb, and H. Huang, "Tuning-free plug-and-play proximal algorithm for inverse imaging problems," 2020, *arXiv:2002.09611*. [Online]. Available: http://arxiv.org/abs/2002.09611

[60] Y. Sun, Z. Wu, X. Xu, B. Wohlberg, and U. S. Kamilov, "Scalable plug-and-play ADMM with convergence guarantees," 2020, *arXiv:2006.03224*. [Online]. Available: http://arxiv.org/abs/2006.03224

Weisheng Dong (Member, IEEE) received the B.S. degree in electronic engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2004, and the Ph.D. degree in circuits and system from Xidian University, Xi'an, China, in 2010.

He was a Visiting Student with Microsoft Research Asia, Beijing, China, in 2006. From 2009 to 2010, he was a Research Assistant with the Department of Computing, The Hong Kong Polytechnic University, Hong Kong. In 2010, he joined the School of Electronic Engineering, Xidian University, as a Lecturer, where he has been a Professor since 2016. His research interests include inverse problems in image processing, deep learning, computer vision, and computational imaging. He was a recipient of the Best Paper Award at the SPIE Visual Communication and Image Processing (VCIP) in 2010. He has served as an Associate Editor for IEEE TRANSACTIONS ON IMAGE PROCESSING and *SIAM Journal of Imaging Sciences*.

Chen Zhou received the B.S. degree in electrical engineering from Xidian University, Xi'an, China, in 2018, where she is currently pursuing the master's degree with the School of Artificial Intelligence. Her research interests include hyperspectral image reconstruction and deep learning.

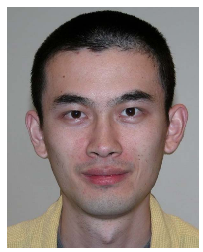
Fangfang Wu received the B.S. degree in electrical engineering from Xidian University, Xi'an, China, in 2008, where she is currently pursuing the Ph.D. degree with the School of Artificial Intelligence.

Her research interests include image restoration, hyperspectral imaging, and deep image representation.

Jinjian Wu (Member, IEEE) received the B.Sc. and Ph.D. degrees from Xidian University, Xian, China, in 2008 and 2013, respectively. From 2011 to 2013, he was a Research Assistant with Nanyang Technological University, Singapore, where he was a Postdoctoral Research Fellow from 2013 to 2014. From 2015 to 2019, he was an Associate Professor with Xidian University, where he has been a Professor since 2019. His research interests include visual perceptual modeling, biomimetic imaging, quality evaluation, and object detection. He received the Best Student Paper Award at ISCAS 2013. He has served as an Associate Editor for the journal of *Circuits, Systems and Signal Processing (CSSP)*, the Special Section Chair for the 2017 IEEE Visual Communications and Image Processing (VCIP), and the Section Chair/Organizer/TPC Member for ICME 2014–2015, PCM 2015–2016, ICIP 2015, VCIP 2018, and AAAI 2019 Quality Assessment.

Guangming Shi (Fellow, IEEE) received the B.S. degree in automatic control, the M.S. degree in computer control, and the Ph.D. degree in electronic information technology from Xidian University in 1985, 1988, and 2002, respectively.

He joined the School of Electronic Engineering, Xidian University, in 1988. From 1994 to 1996, as a Research Assistant, he cooperated with the Department of Electronic Engineering, The University of Hong Kong. Since 2003, he has been a Professor with the School of Electronic Engineering, Xidian University, and in 2004 the Head of the National Instruction Base of Electrician and Electronic (NIBEE). From June 2004 to December 2004, he studied at the Department of Electronic Engineering, University of Illinois at Urbana-Champaign (UIUC). He is currently the Deputy Director of the School of Electronic Engineering, Xidian University, and the Academic Leader in the subject of circuits and systems. He has authored or coauthored over 100 research articles. His research interests include compressed sensing, theory and design of multirate filter banks, image denoising, low-bit-rate image/video coding, and implementation of algorithms for intelligent signal processing (using DSP and FPGA).



Xin Li (Fellow, IEEE) received the B.S. degree (Hons.) in electronic engineering and information science from the University of Science and Technology of China, Hefei, China, in 1996, and the Ph.D. degree in electrical engineering from Princeton University, Princeton, NJ, USA, in 2000.

He was a Technical Staff Member with the Sharp Laboratories of America, Camas, WA, USA, from 2000 to 2002. Since 2003, he has been a Faculty Member with the Lane Department of Computer Science and Electrical Engineering. His research interests include image/video coding and processing. He was a recipient of the Best Student Paper Award at the Conference of Visual Communications and Image Processing in 2001, the Best Student Paper Award at the IEEE Asilomar Conference on Signals, Systems and Computers in 2006, and the Best Paper Award at the Conference of Visual Communications and Image Processing in 2010. He is also serving as a member for the Image, Video and Multidimensional Signal Processing Technical Committee and an Associate Editor for the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY.